Does the Order of Training Data Influence Optimization?

Rapport_V3: Optimization Dynamics Under Non-IID Training Streams

Niels Enkaoua Teammate 1 Teammate 2

2026-01-01

Abstract

The original project established that data order alone can strongly affect both convergence and generalization. Rapport_V3 extends that study into an optimization analysis. We instrument SGD to measure successive gradient cosine similarity, class-wise accuracy, catastrophic forgetting, and final representation geometry. The setup remains controlled: Fashion-MNIST, a small CNN, SGD with momentum, 10 epochs, a stratified 20,000-example training subset, and seeds 0, 1, and 2. Three conclusions emerge. First, random reshuffling remains the strongest baseline, while fixed random order is consistently slightly worse. Second, the pathological class-ordered streams are not only inaccurate; they induce highly aligned early gradients, severe class imbalance, and degenerate or even non-finite representations. Third, reducing the learning rate partly rescues the class-ordered runs, whereas increasing it leaves random reshuffling relatively robust but preserves collapse for the structured label orders. The resulting picture is that data ordering changes both the noise structure seen by SGD and the stability region of the optimizer.

Introduction

SGD is usually analyzed and implemented under a near-iid sampling assumption: each mini-batch is treated as a noisy but broadly representative estimate of the full gradient (Bottou, Curtis, and Nocedal 2018). In practical pipelines this assumption is approximated by random reshuffling at every epoch, which is known to behave differently from reusing a fixed order (Gürbüzbalaban, Ozdaglar, and Parrilo 2021). When the training stream becomes structured, however, consecutive updates need not point in diverse directions. They may instead become strongly correlated, concentrate on a narrow subset of classes, and accumulate bias through momentum.

This project asks a deliberately narrow question: if the dataset, model, optimizer, and seeds are fixed, does changing only the training-data order alter optimization dynamics and final generalization? Earlier project versions already showed that random reshuffling is best, fixed random order is slightly worse, curriculum-hard is better than curriculum-easy, and class-ordered streams can collapse to chance-level accuracy. Rapport_V3 strengthens that story by measuring the dynamics behind those outcomes rather than only the final test accuracies.

Why Gradient Correlation Matters

The relevant quantity is not just gradient magnitude but gradient direction. If consecutive stochastic gradients are weakly correlated, SGD keeps sampling different local directions and the resulting noise acts as a regularizing perturbation. If consecutive gradients are strongly aligned, momentum reinforces the same directional bias across many steps. Under a class-ordered stream this means that the optimizer can drift toward a decision rule specialized to the currently visible class block instead of maintaining a balanced classifier. In that sense, the order of the stream changes the temporal correlation structure of the noise injected into SGD.

This motivates the first new diagnostic: after every backward pass, we concatenate all parameter gradients into one vector and compute the cosine similarity between consecutive steps. High cosine similarity indicates persistent directional alignment; low or sign-changing cosine similarity indicates more diverse stochastic updates. The central hypothesis is that random reshuffling should decorrelate neighboring gradients, whereas label-based orders should create long aligned sequences and therefore a much more brittle optimization trajectory.

Experimental Setup

The dataset is Fashion-MNIST (Xiao, Rasul, and Vollgraf 2017). The model is the same compact CNN used in the previous report, with two convolutional blocks and a two-layer classifier, and without BatchNorm or Dropout. The optimizer is SGD with momentum 0.9, weight decay $5 \cdot 10^{-4}$, batch size 128, and the main learning rate 0.05. All reported main runs use 10 epochs, seeds 0, 1, and 2, and a stratified subset of 20,000 training examples.

We compare six orderings: `random`, `fixed_random`, `curriculum_easy`, `curriculum_hard`, `label_sorted`, and `label_block_random`. In addition to train loss and test accuracy, Rapport_V3 records:

- gradient cosine similarity at every optimization step,
- per-class test accuracy after every epoch,
- a forgetting statistic derived from the best-so-far class accuracy,
- final predicted-class distributions,
- and final hidden-layer embeddings on the test set for PCA visualization.

For learning-rate sensitivity, the V3 code supports a sweep over 0.1, 0.05, 0.01, and 0.001. In the runs executed for this report, the new high-learning-rate diagnostic at 0.1 was completed for the principal orderings, and the low-learning-rate rescue runs at 0.001 were available for the two pathological label-based streams.

Main Ordering Effects

The main ranking remains clear. Random reshuffling reaches the best final accuracy (89.47%), fixed random order is slightly worse (88.46%), and curriculum-hard (84.53%) remains meaningfully better than curriculum-easy (70.71%). By contrast, both label-based streams collapse to 10% accuracy, which is exactly chance for a balanced ten-class test set. Relative to the previous report, the main message is therefore stable: reshuffling between epochs helps, and strongly non-iid streams are dangerous even when everything else is held fixed.

The corresponding trajectories and final summary are reported in Appendix A.

Gradient Cosine Similarity

The cosine diagnostic sharpens the optimization story. Random reshuffling and fixed random order both produce low average alignment, around 0.20. The two curriculum variants are slightly more correlated, around 0.25. The most striking case is `label_block_random`, whose valid cosine values average 0.92, indicating that successive updates point in almost the same direction for the part of training that remains numerically finite. `label_sorted` begins with the same qualitative behavior, but it exits the stable regime extremely quickly: only 42 of 4710 recorded step-to-step pairs remain finite, and the last finite cosine value appears by step 80. In other words, the fully sorted order does not merely yield correlated gradients; it drives the optimizer into numerical breakdown almost immediately.

This is exactly the mechanism expected from non-iid blocks. Consecutive batches coming from the same or closely related labels induce highly aligned updates, and momentum accumulates that alignment rather than averaging it away. Random reshuffling, in contrast, keeps changing the local direction and therefore prevents sustained drift toward a single-class solution.

The full time-series plot is provided in Appendix B.

Per-Class Accuracy and Forgetting

The class-wise diagnostics reveal what the aggregate accuracy hides. Under random reshuffling, the final classifier is reasonably balanced across classes. Under `label_sorted`, however, the final model is entirely degenerate: class 0 reaches 100% accuracy while every other class remains at 0%. Under `label_block_random`, only a small subset of classes retains any non-zero accuracy, and the dominant class changes across epochs and seeds. This is consistent with catastrophic forgetting in the broad optimization sense: information about previously encountered class blocks is not retained once later blocks dominate the update stream (McCloskey and Cohen 1989).

The forgetting curve must be interpreted carefully. For `label_block_random`, mean forgetting rises steadily and reaches 0.17 by epoch 10, which matches the intuition that temporarily acquired class performance is later overwritten. For `label_sorted`, the forgetting statistic alone is less informative because most classes are never learned at all after the early numerical breakdown. In that case the heatmap is the more faithful diagnostic: the stream does not merely forget old classes, it fails to build a balanced multi-class representation in the first place.

The per-class heatmaps, representative class trajectories, and forgetting curve are grouped in Appendix C.

Learning-Rate Sensitivity

The learning-rate study reinforces the main conclusion that stability depends on order. Random reshuffling remains robust when the learning rate is increased from 0.05 to 0.1, dropping only from 89.47% to 87.96%. Fixed random order degrades slightly from 88.46% to 86.75%, and curriculum-hard degrades more substantially from 84.53% to 79.52%, but these runs remain trainable. The class-ordered streams do not. Both `label_sorted` and `label_block_random` stay collapsed at 10% accuracy for both 0.05 and 0.1, and their final train losses become extremely large. Lowering the learning rate to 0.001 partially rescues these pathologies, improving `label_sorted` to 31.95% and `label_block_random` to 40.00%, but both remain far below the random baseline.

This is the clearest evidence that data order changes the stability region of SGD. The same optimizer and architecture tolerate a larger step size under reshuffled data, while the structured label orders operate in a much narrower regime. High correlation between consecutive gradients makes the effective momentum-driven drift stronger, so increasing the learning rate amplifies the bias rather than simply accelerating learning. Lowering the step size can reduce this amplification, but it does not restore the diversity of directions produced by reshuffling.

The learning-rate sweep figures are collected in Appendix D.

Representation Geometry

The PCA visualizations are consistent with the optimization diagnostics. Random reshuffling, fixed random order, and curriculum-hard produce recognizable class structure in the final hidden representation. The label-based runs do not. For the collapsed orders the final embedding files contain non-finite activations, so the corresponding panels show a numerical failure rather than a meaningful geometric arrangement. This is useful evidence in itself: the failure is not confined to the classifier head or to a single metric, but propagates into the learned representation.

The full PCA panel appears in Appendix E.

Discussion

Rapport_V3 changes the interpretation of the project from a simple benchmark into a dynamical study. The baseline result, that random reshuffling is best, is still true, but the new diagnostics explain why. Random reshuffling reduces temporal correlation between successive stochastic gradients and keeps the optimization process in a broad, stable regime. Fixed random order removes the benefit of epoch-to-epoch reshuffling, which is enough to produce a small but consistent drop. Curriculum-hard and curriculum-easy confirm that deterministic order does matter even outside the pathological class-sorted case, and that the effect depends on the information content of the early stream.

The class-ordered failures are the central phenomenon. They produce highly aligned early gradients, severe class imbalance in test predictions, rising forgetting for the partially trainable block-random variant, and degenerate or non-finite final representations. The high-learning-rate runs show that this is a genuine stability issue rather than a harmless slowdown. The low-learning-rate rescue runs show that step size matters, but also that simply reducing the learning rate does not recover the behavior of random reshuffling.

Conclusion

The order in which training examples are presented changes SGD in a mechanically meaningful way. It alters gradient correlation, class retention, representation quality, and the range of learning rates for which training remains stable. Random reshuffling is not just a convenience; in this controlled Fashion-MNIST setting it is the most robust way to maintain diverse stochastic updates and avoid collapse. Highly structured non-iid streams can instead drive SGD toward biased, forgetful, or numerically unstable behavior. The main lesson is therefore optimization-theoretic: data order changes the noise model seen by SGD, and that change is large enough to affect both convergence and generalization.

Reproducibility

Main V3 diagnostics at the reference learning rate:

```
python ..\run_v3.py --results-root results_v3 --main-output-dir results_v3_main --skip-sweep --epochs 10 -
```

High-learning-rate extension used in this report:

```
python ..\run_v3.py --results-root results_v3 --main-output-dir results_v3_main --skip-main --sweep-learni
```

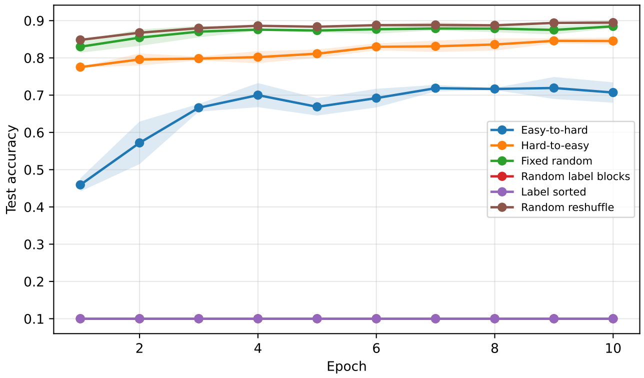
Full V3 sweep pipeline, including the extra 0.01 and 0.001 runs:

```
python ..\run_v3.py --results-root results_v3 --main-output-dir results_v3_main --epochs 10 --seeds 0 1 2
python ..\run_v3.py --results-root results_v3 --main-output-dir results_v3_main --plot-only --extra-sweep-
```

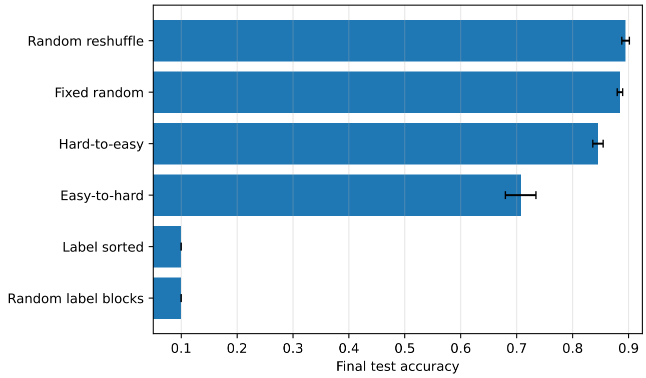
Render the report from inside `Rapport_V3/`:

```
quarto render Rapport_V3.qmd --to pdf
```

Appendix A: Main Ordering Figures

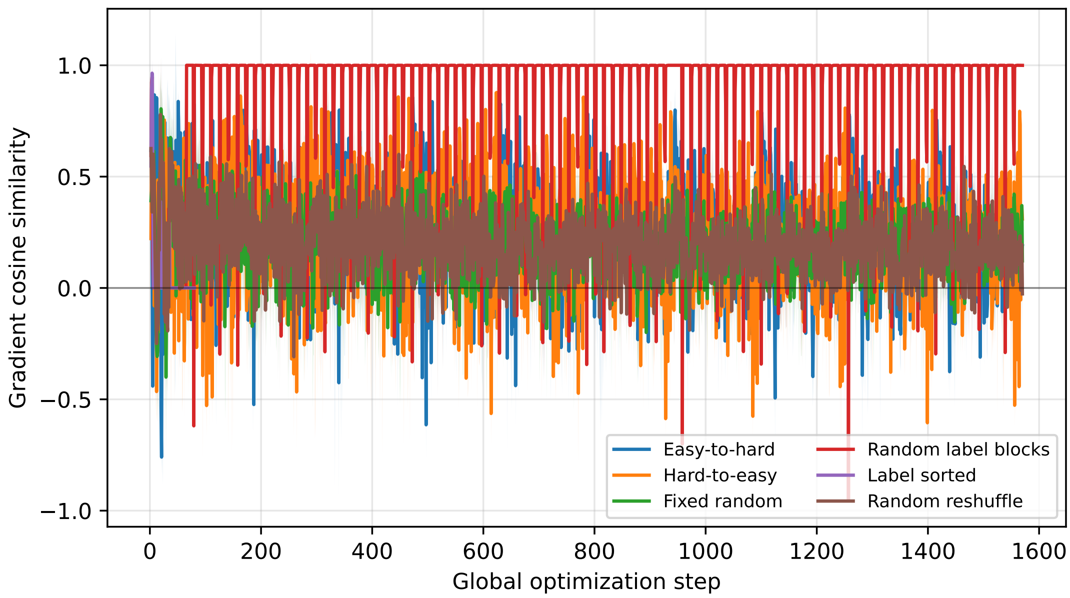


Main test accuracy trajectories at $lr = 0.05$.



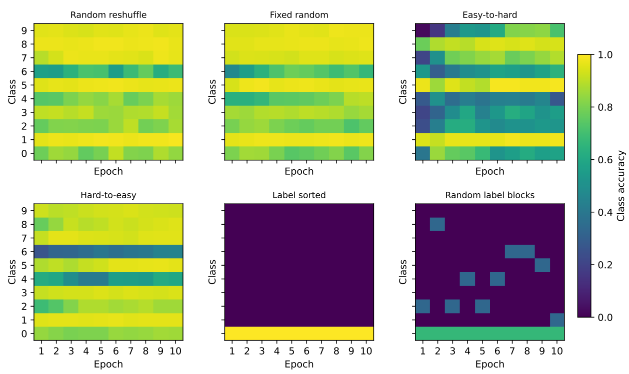
Final test accuracy, mean $\pm$ std across seeds.

Appendix B: Gradient Cosine Similarity

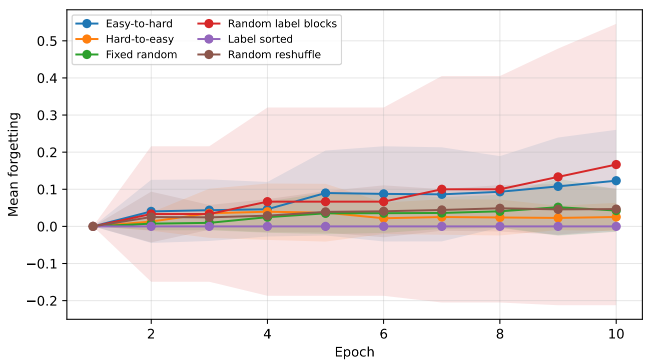


Successive gradient cosine similarity over optimization steps.

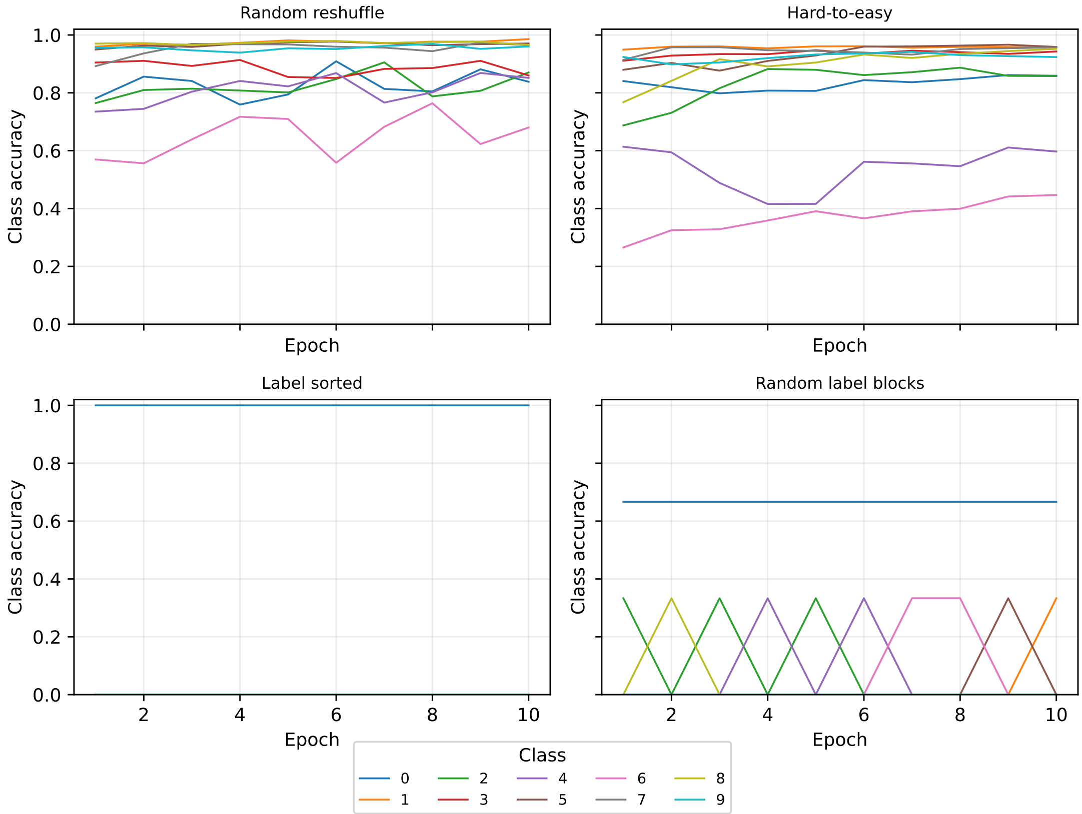
Appendix C: Class Accuracy and Forgetting



Per-class test accuracy heatmaps across epochs.

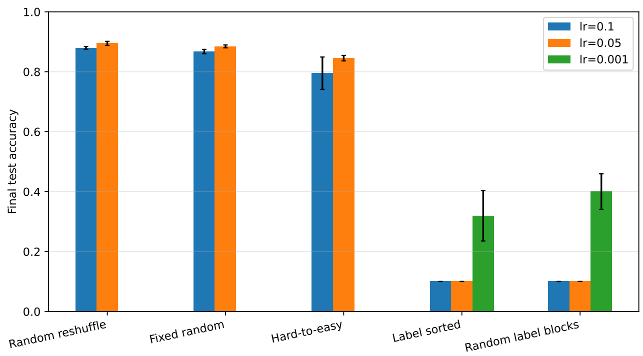


Mean forgetting across classes and seeds.

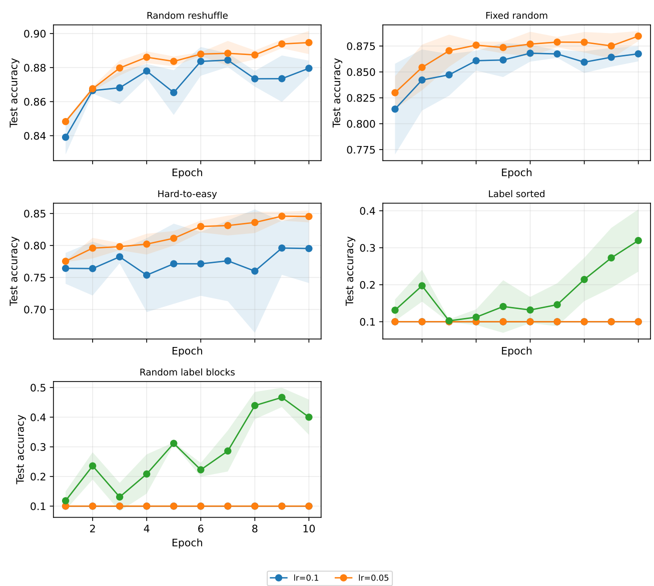


Class-wise accuracy trajectories for representative orders.

Appendix D: Learning-Rate Sensitivity

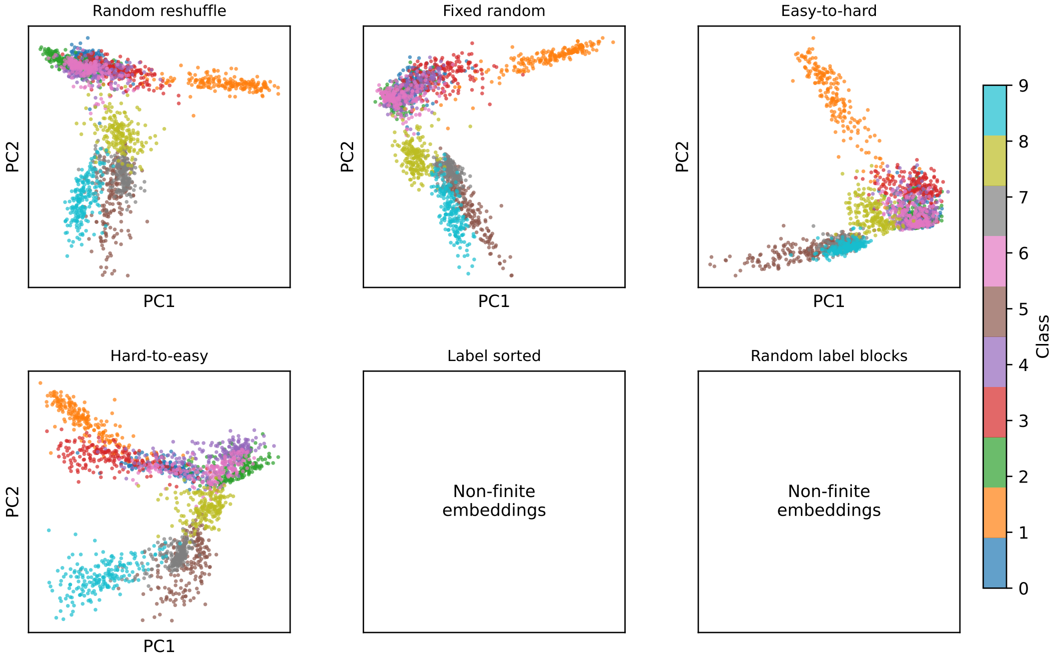


Final accuracy across the executed learning-rate sweep.



Accuracy trajectories for the principal orderings.

Appendix E: PCA Projections



PCA projections of final hidden-layer embeddings on the test set.

References

Bottou, Léon, Frank E. Curtis, and Jorge Nocedal. 2018. "Optimization Methods for Large-Scale Machine Learning." *SIAM Review* 60 (2): 223–311.

Gürbüzbalaban, Mert, Asuman Ozdaglar, and Pablo A. Parrilo. 2021. "Why Random Reshuffling Beats Stochastic Gradient Descent." *Mathematical Programming* 186: 49–84.

McCloskey, Michael, and Neal J. Cohen. 1989. "Catastrophic Interference in Connectionist Networks: The Sequential Learning Problem." *Psychology of Learning and Motivation* 24: 109–65.

Xiao, Han, Kashif Rasul, and Roland Vollgraf. 2017. "Fashion-MNIST: A Novel Image Dataset for Benchmarking Machine Learning Algorithms." *arXiv Preprint arXiv:1708.07747*.